

JKBOSE

2026 CLASS 11TH

MOTION IN STRAIGHT LINE

LECTURE-4

PHYSICS

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Sir



Topics To Be Covered

- 1 SLOPE OF $x-t$ graph ✓ → Board | Neet | Jee
- 2 Area of $v-t$ graph ✓ → Board | Neet | Jee
- 3 $v = u + at$ (derivation)
- 4 $s = ut + \frac{1}{2} at^2$ (D)
- 5 $v^2 = u^2 + 2as$ (D)

“DON'T WAIT FOR THE
PERFECT MOMENT,
TAKE THE MOMENT AND
MAKE IT PERFECT”

Recap From 1st Lec.

Motion

- move one place to another place with respect to

Rest

Donot move from one place to another place with respect to

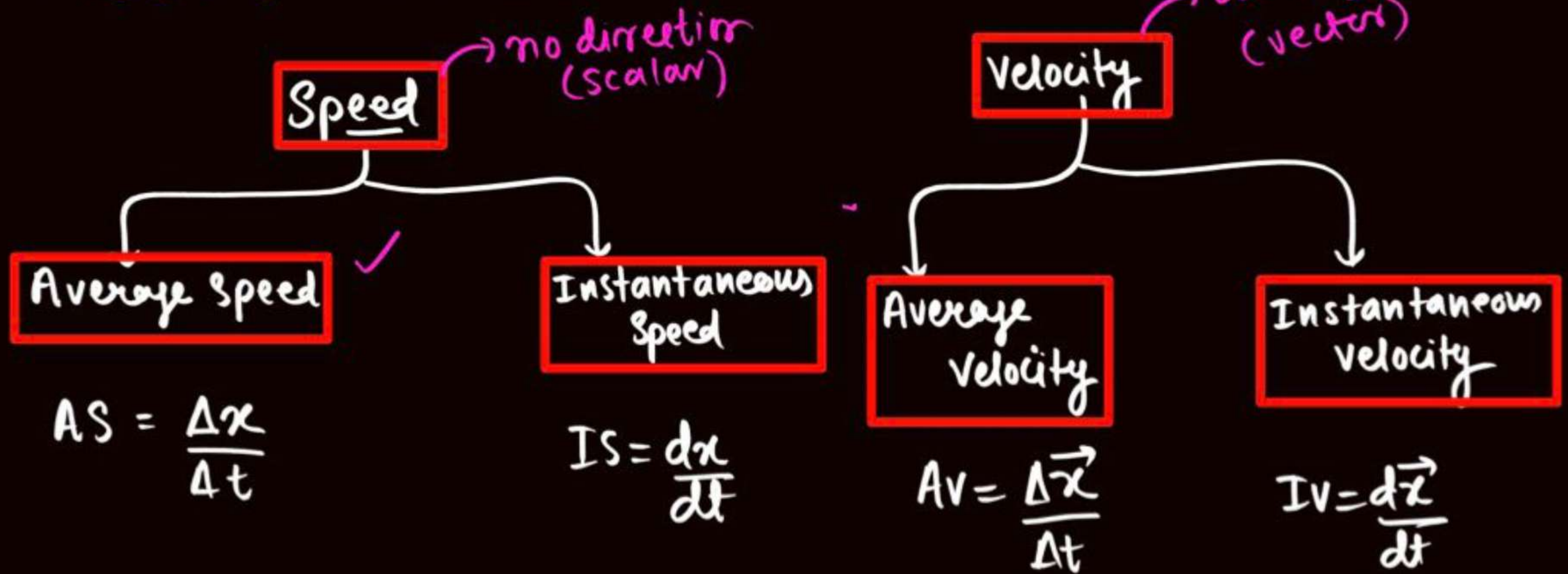
Reference point
(Motion is Relative)

$$1 \frac{\text{km}}{\text{hr}} = \frac{5}{18} \text{ m/s}$$

Numericals

distance	displacement
• Total path • metre • Scalar • Positive	• Shortest path • metre • vector + - zero
Speed	Velocity
• $\text{Speed} = \frac{\text{distance}}{\text{time}}$ • Scalar • m/s time	• $\text{Velocity} = \frac{\text{displacement}}{\text{time}}$ • Vector - m/s • +ive -ive 0

Recap 2nd lecture



Lec. 3rd Recap.

Acceleration

⇒ Rate of change of velocity

$$a = \frac{\Delta v}{t} \quad \# \text{ Vector} \quad \# \text{ m/s}^2$$

⇒ $a = +ve$, $a = 0$, $a = -ve$

Retardation ♡

Motion

Uniform Motion

$$a = 0$$

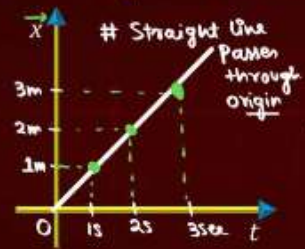
Non-uniform Motion

$$a > 0$$

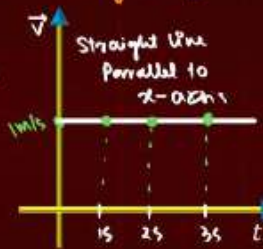
$$a < 0$$

Graphs

① displacement-time graph



② velocity-time graph

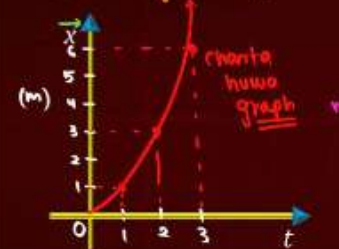


acceleration-time graph

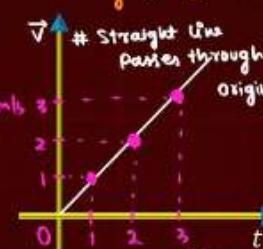


Graphs

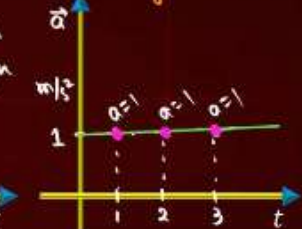
① displacement-time graph



② velocity-time graph

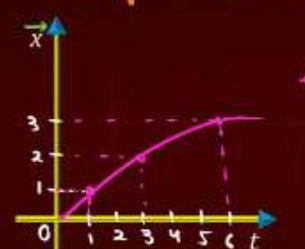


acceleration-time graph



Graphs

① displacement-time graph



② velocity-time graph



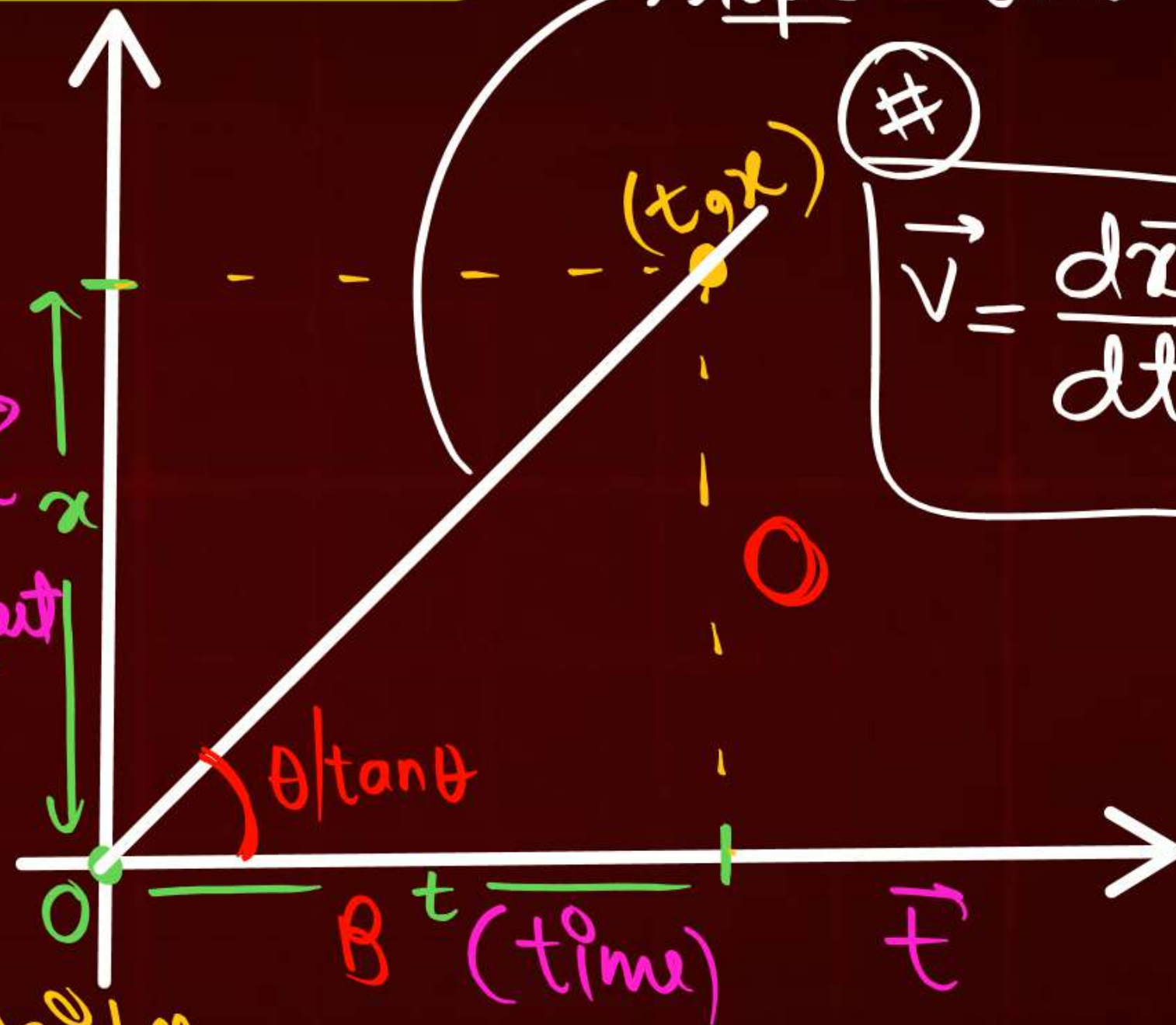
acceleration-time graph



SLOPE OF $\vec{x}t$ graph gives

- (a) acceleration
- (b) ✓ velocity
- (c) acceleration
- (d) All the above

$$\text{Slope} = \tan \theta = \frac{O}{B} = \frac{\vec{x}}{t} = \text{velocity}$$



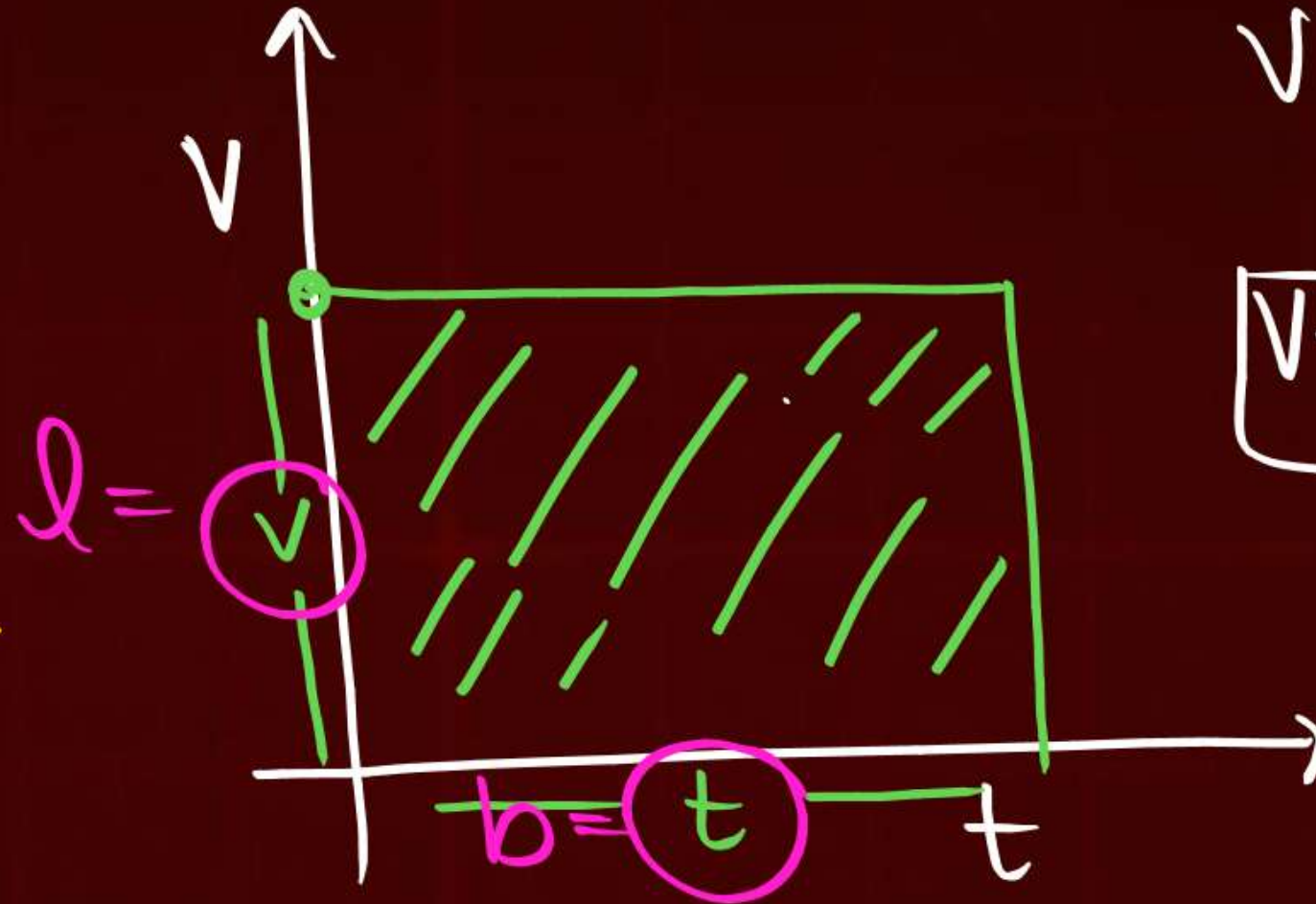
$$\text{Slope} = \tan \theta = \frac{d\vec{x}}{dt}$$

$$\vec{v} = \frac{d\vec{x}}{dt}$$

Area of vt graph

(Neet-2019)

- (a) displacement
- (b) acceleration
- (c) velocity
- (d) All of above



$$v = \frac{x}{t}$$

$$vt = x$$

Area under vt = $L \times B = \underline{\underline{\vec{v} \times t}} = \underline{\underline{\vec{x}}}$

Q. 1

A person travel x -distance with velocity v_1 and x -distance with v_2 in the same direction the average velocity (v) of a person in term of v_1 and v_2

Jee-2023

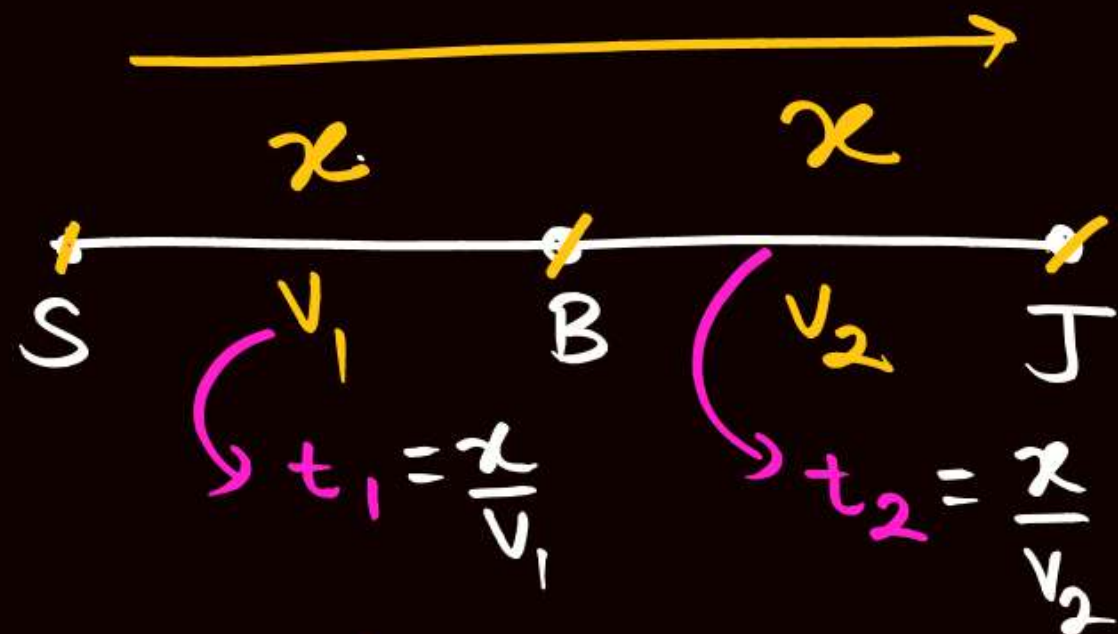
(a) $V = v_1 + v_2$

(b) $V = \frac{v_1^2 + v_2^2}{2}$

(c) $\frac{2}{V} = \frac{1}{v_1} + \frac{1}{v_2}$

(d) $\frac{1}{V} = \frac{1}{v_1} + \frac{1}{v_2}$

Sol



$$V = ?$$

$$Td = x + x = 2x$$

$$Tt = t_1 + t_2$$

Sol

$$V_a = \frac{Td}{Tt}$$

$$V_a = \frac{2x}{t_1 + t_2}$$

$$V = \frac{2x}{\frac{x}{v_1} + \frac{x}{v_2}}$$

$$V = \frac{2x}{x \left(\frac{1}{v_1} + \frac{1}{v_2} \right)}$$

$$\frac{V}{1} = \frac{2 \times 1}{\left(\frac{1}{v_1} + \frac{1}{v_2} \right)}$$

$$\left(\frac{V}{2} \right) = \left(\frac{1}{\left(\frac{1}{v_1} + \frac{1}{v_2} \right)} \right)$$

$$V = \frac{x}{t} \rightarrow t_2 = \frac{x}{v_2}$$
$$t = \frac{x}{V}$$
$$t_1 = \frac{x}{v_1}$$

$$\frac{2}{V} = \left(\frac{1}{v_1} + \frac{1}{v_2} \right)$$

$$\boxed{\frac{2}{V} = \frac{1}{v_1} + \frac{1}{v_2}}$$

Newton equation of motion

There are 3 equation
of motion

$$\textcircled{1} \quad v = u + at \quad \checkmark$$

$$\textcircled{2} \quad s = ut + \frac{1}{2}at^2 \quad \checkmark$$

$$\textcircled{3} \quad v^2 = u^2 + 2as \quad \checkmark$$

✓ $v \rightarrow$ Final Velocity

✓ $u \rightarrow$ initial velocity

✓ $a \rightarrow$ acceleration

✓ $t \rightarrow$ time

✓ $s \rightarrow$ displacement

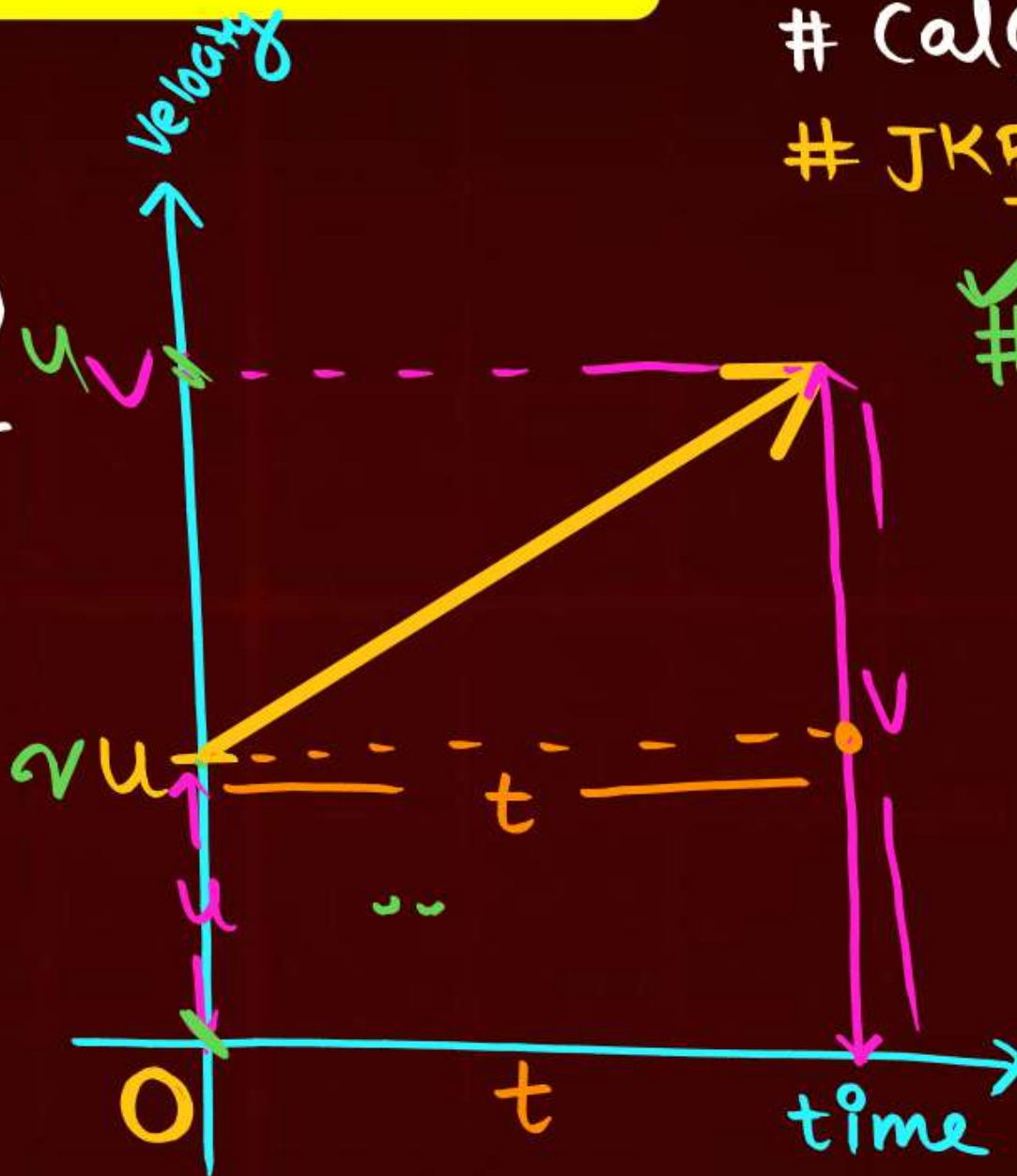
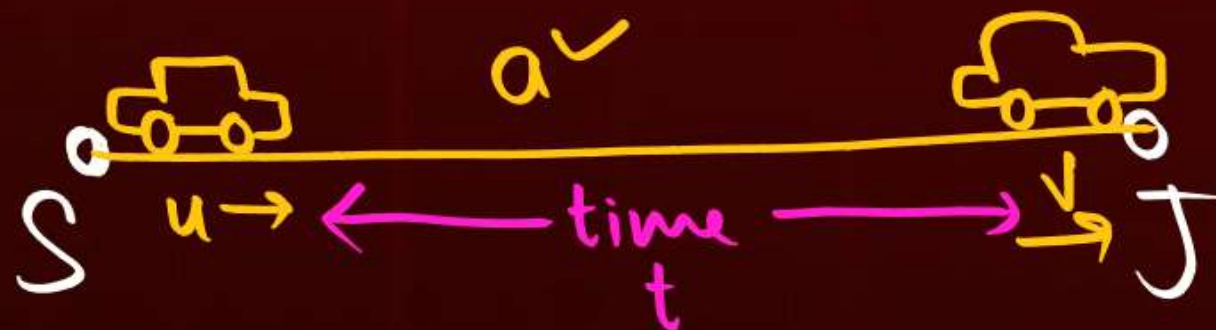
Derivative of equation of Motion

→ # Graphical Method ✓



(i) $v = u + at$

- Let the Body (Car of Shagufta) moves with initial velocity (u) and reach the point with Final velocity (v) having acceleration (a) for the time period (t)



Calculus X

2025 2023
2024 2021

JKBOSE → 5 marks

✓ # $a = \frac{v_f - v_i}{t}$

$a = \frac{v - u}{t}$

$at = v - u$

$u + at = v$

✓ $v = u + at$

Note ① $\check{V} = \check{U} + \check{a}t$

displacement $s = x$

Numerical

Q A Shagufa car moves with initial velocity ($U=0$)
having acceleration equal to 2m/s^2 for a time period of
2Sec calculate the $V=?$

Sol

$$V = U + at$$

$$V = 0 + 2 \times 2$$

$$V = 0 + 4$$

$$\underline{V = 4\text{m/s}}$$

V ✓

a ✓

t ✓

u ✓

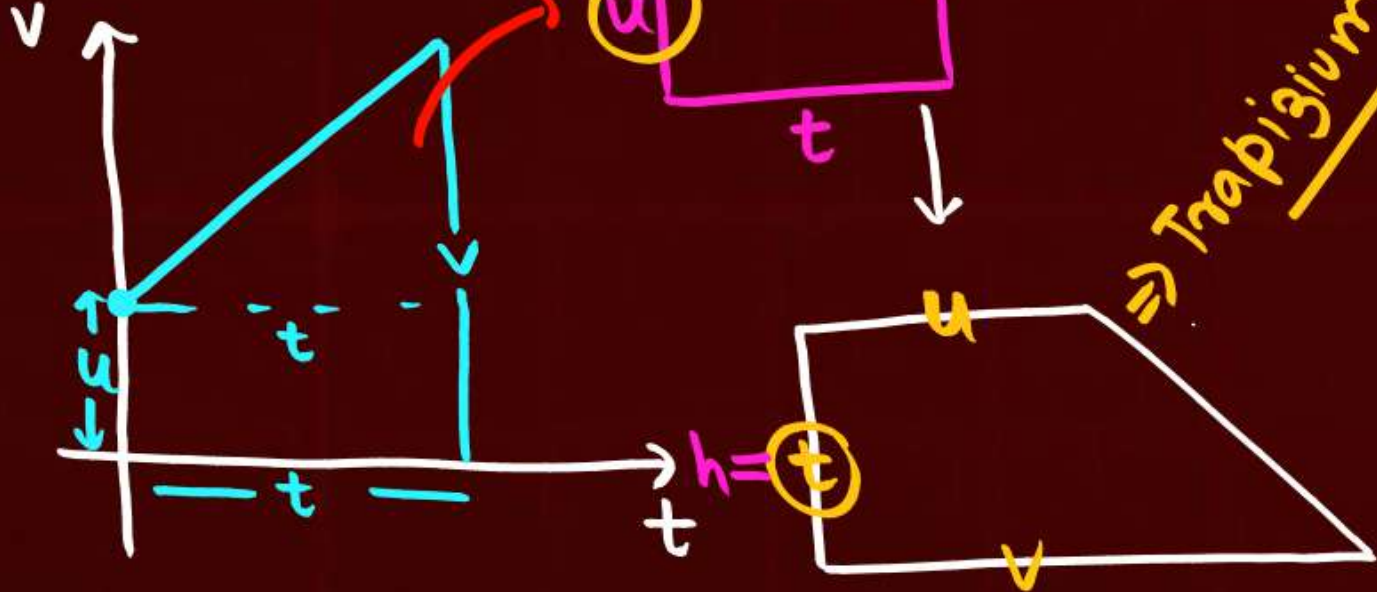
Derivative of equation of Motion

$$\frac{x}{2} = \frac{1}{2}x \quad \checkmark$$



(ii) $s = ut + \frac{1}{2}at^2$

Graph



displacement = Area under vt-graph

$S = \text{Area of Trapezium}$

$$S = \frac{1}{2} (\text{sum of || Side}) \times \text{height}$$

$$S = \frac{1}{2} (u + v) \times t$$

$$v = u + at$$

$$S = \frac{1}{2} (u + u + at) \times t$$

$$S = \frac{t}{2} (2u + at)$$

$$S = \cancel{u}t \frac{2}{2} + at \frac{t}{2}$$

$$S = ut + \frac{at^2}{2}$$

$$S = ut + \frac{1}{2}at^2$$

Note :-

(2)

$$S = \check{u} \check{t} + \frac{1}{2} \check{a} \check{t}^2$$

$$\check{v}^x$$

Numerical

Q. A Laadale starts his car with initial velocity $u=0$ having $a=2\text{m/s}^2$ for the time period of 10sec. Calculate the displacement travel by laadale car?

Sol. (a) 10m

☒ (b) 100m

(c) 1000m

(d) 10,000m

$$S = ut + \frac{1}{2}at^2$$

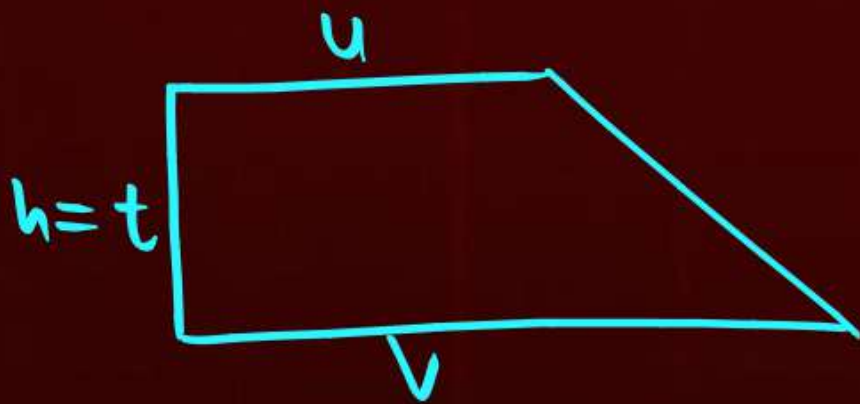
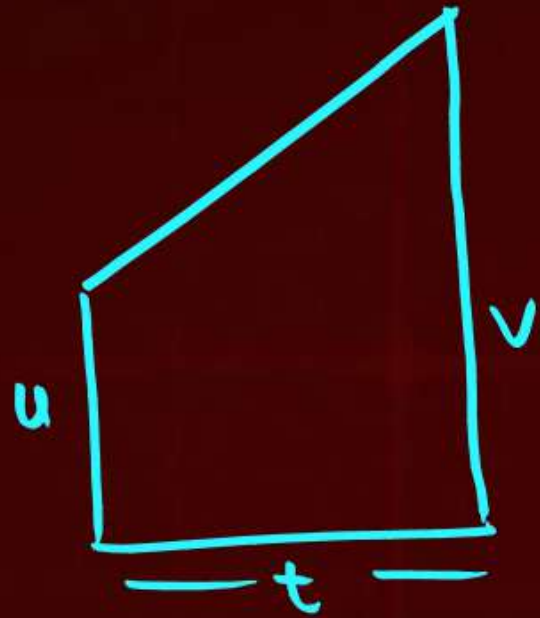
$$S = 0 \times 10 + \frac{1}{2} \times 2 \times (10)^2$$

$$S = 0 + (10)^2$$

$$S = 10^2 = 100\text{m}$$

Derivative of equation of Motion

(iii) $v^2 = u^2 + 2as$ $t=x$ Displacement = area under vt-graph.



S = area of trapezium

$$S = \frac{1}{2} (\text{sum of || side}) \times h$$

$$S = \frac{1}{2} (u+v) \times t$$

$$v = u + at$$

$$v - u = at$$

$$\frac{v-u}{a} = t$$

$$S = \frac{1}{2} (u+v) \frac{(v-u)}{a}$$

$$2aS = (u+v)(v-u)$$

$$2as = (u+v)(v-u)$$

$$2+4=6$$

$$4+2=6$$

$$2as = (v+u)(v-u)$$

$$(a+b)(a-b) = a^2 - b^2$$

$$2as = v^2 - u^2$$

$$u^2 + 2as = v^2$$

$$v^2 = u^2 + 2as$$

Note:- (3)

$$\underline{v^2 = u^2 + 2as}$$

v ✓

u ✓

a ✓

s ✓

$t \times$

Q A daadale starts his car with initial velocity ($u=0$)
travel a displacement of 100m with an acceleration
of 2m/s^2 . Calculate the final velocity attain by
daadale?

$$u=0$$

$$a=2$$

$$s=100\text{m}$$

$$v=?$$

Sol

$$v^2 = u^2 + 2as$$

$$v^2 = 0^2 + 2 \times 2 \times 100$$

$$v^2 = 0 + 400$$

$$v^2 = 20^2$$

$$v = \sqrt{20^2} = 20\text{m/s} \quad \checkmark$$

$$(a) \ 10\text{m/s}$$

$$(b) \ 20\text{m/s}$$

$$(c) \ 30\text{m/s}$$

$$(d) \ 40\text{m/s}$$

(g) Acceleration due to gravity (g) Gravitational field



↓ qth ✓

Gravitation $g = \frac{GM}{R^2}$ (11th)

the acc. caused by
the earth is
called acc. due to
gravity



$$F = ma$$
$$F = mg$$

$$a = g = -9.8 \text{ m/s}^2$$
$$= -10 \text{ m/s}^2$$



numericals

Equation of Motion (gravit)
 $a = -g$

$$v = u + at$$

$$v = u + -gt$$

$$v = u - gt$$

$$s = ut + \frac{1}{2}at^2$$

$$s = ut + \frac{1}{2}-gt^2$$

$$s = ut - \frac{1}{2}gt^2$$

$$v^2 = u^2 + 2as$$

$$v^2 = u^2 + 2(-g)s$$

$$v^2 = u^2 - 2gs$$

Chocolate Question [2025-B]

Which is the formula for motion in straight line?

1 $v = u + at$ ✓

2 $v = 2at + v$

3 $v = u - at$

4 $v = 2at + u$

Chocolate Question [2025-C] ✓

From the velocity time graph of uniform accelerated motion, deduce the equation of motion in distance and time. ✓

$$v = u + at \quad ✓$$

$$s = ut + \frac{1}{2}at^2 \quad ✓$$

$$v^2 = u^2 + 2as \quad ✓$$

Chocolate Question [2025-C]

Which among the following can be zero when a particle is in motion for some time?



- 1 Displacement = 0 ✓
- 2 Speed = $\frac{d}{t} = \frac{200m}{2sec} = 100m/s$
- 3 Distance = $100 + 100 = 200m$
- 4 None of these ✓

SUMMARY



Slope of x vs t

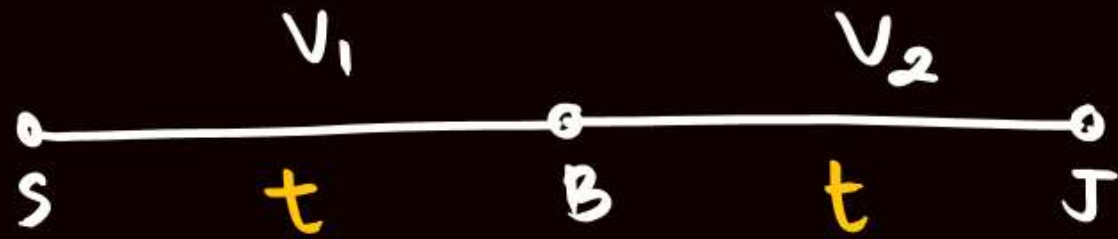
Area of v vs t

$$\# \quad v = u + at$$

$$s = ut + \frac{1}{2}at^2$$

$$v^2 = u^2 + 2as$$

H/W



$$(i) \frac{2}{V} = \frac{1}{V_1} + \frac{1}{V_2}$$

$$(ii) \frac{V_1^2 + V_2^2}{2}$$

$$(iii) \frac{V_1 + V_2}{2}$$

$$(iv) \sqrt{\frac{V_1 + V_2}{2}}$$

$$V = 9$$

Boared

JKcet-2011

$$(R_s = 100 \Omega) \checkmark$$

Thank
You